Large Language Models

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Most slides have been adopted from LLM Sharif course,
Dr. Asgari, Dr. Rohban, Soleymani, Sharif University of Technology, Fall 2023

What is a language model?

- A probabilistic model that assigns a probability $P(w_1, w_2, \ldots, w_n)$ to every finite sequences of tokens

- Generation from a language model: $x_{1:L} \sim p$

Autoregressive language models

$$P(x_1, \dots, x_T) = P(x_1) \prod_{t=2}^{T} P(x_t | x_1, \dots, x_{t-1})$$

- $P(x_t | x_{1:t-1})$ is modeled efficiently (e.g., using a feedforward NN)

- Example:

$$P(He\ wants\ to\ know\ it)$$

$$= P(He)P(wants|He)P(to|He\ wants)P(know|He\ wants\ to)P(it|He\ wants\ to\ know)$$

Autoregressive language models: Generation

- Generation: sample one token at a time given the tokens generated so far:

for i=1,…,L:

$$x_i \sim p_T(x_i|x_{1:i-1})$$

- $p_T(x_i|x_{1:i-1}) \propto p(x_i|x_{1:i-1})^{1/T}$ is an annealed conditional probability distribution
- $T \geq 0$ controls randomness

Autoregressive language models

- **Conditional generation**. Specifying a prefix $x_{1:i}$, called a **prompt**, and sampling the rest $x_{i+1:L}$
- For example, generating with T=0 produces

Prompt: The, dog, eats
Completion (T=0): the bone

- Conditional generation unlocks the ability to solve a variety of tasks by simply changing the prompt.

History: N-gram language modeling

- n-gram models: $p(w_1, w_2, \ldots, w_n)$ is computed based on the number of times various n-grams occur
- computationally efficient and statistically inefficient.
  - If n is too big, it will be statistically infeasible to find good estimates
- Applications: speech recognition, machine translation, spelling correction,…
- Useful for short context lengths and so employ along with another model (acoustic model or translation model).

History: Neural language models

- Bengio et al. proposed neural language models in 2003:

$$NN(w_1, \ldots, w_n) \approx p(w_n | w_1, \ldots w_{n-1})$$

distributed feature vectors

- Neural language models are statistically efficient but computationally inefficient
  - Their training was not scalable

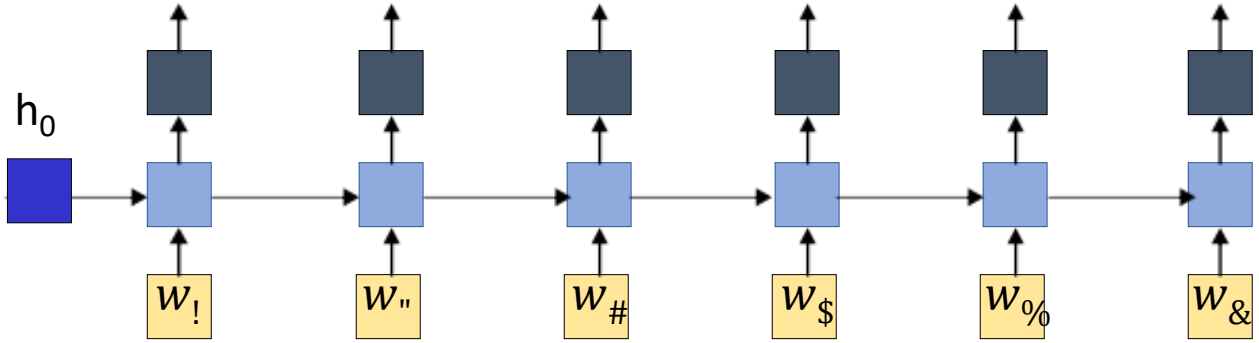
"The cat is walking in the bedroom"

"A dog was running in a room"

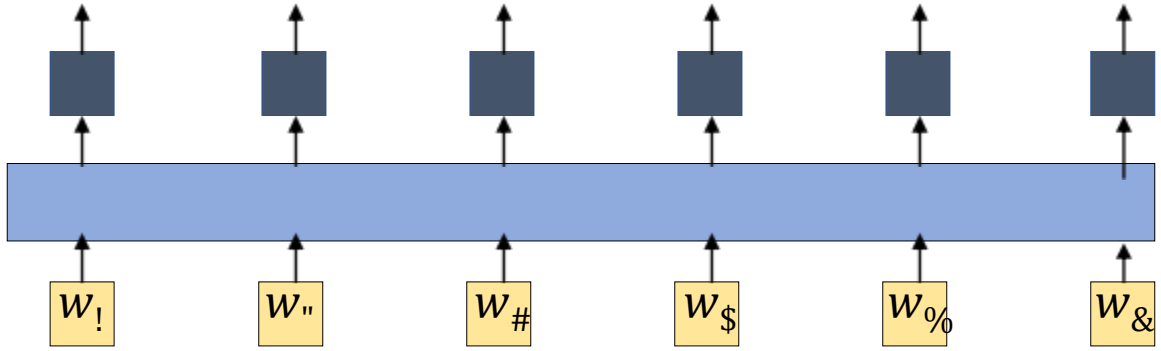
Bengio et al., 2003

History: Neural architectures for language modeling

- Recurrent Neural Networks (RNNs)
  - to depend on the entire context $x_{1:i-1}$



- Transformers (developed for translation in 2017) are much easier to train and exploited the parallelism of GPUs although returned to having fixed context length n
  - GPT3 uses $n = 2048$



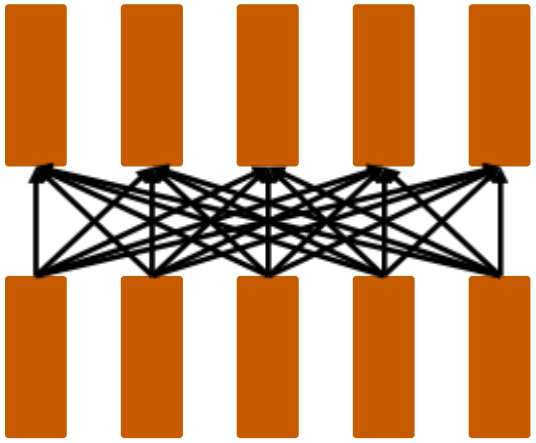
- Neural language models have become the dominant paradigm

History: How Large Language Models (LLMs)?

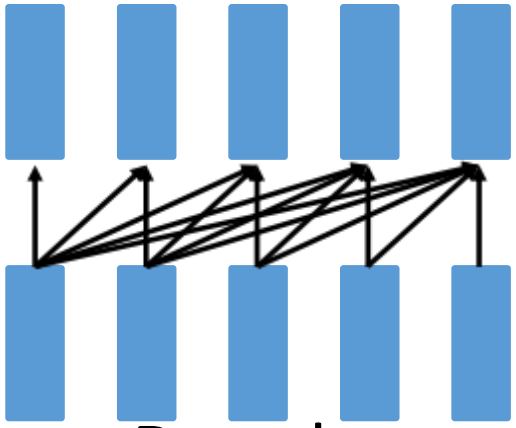
- Hardware improvements
- Transformer model architecture
- Data availability
- Self-supervised approach of pretraining

Language models

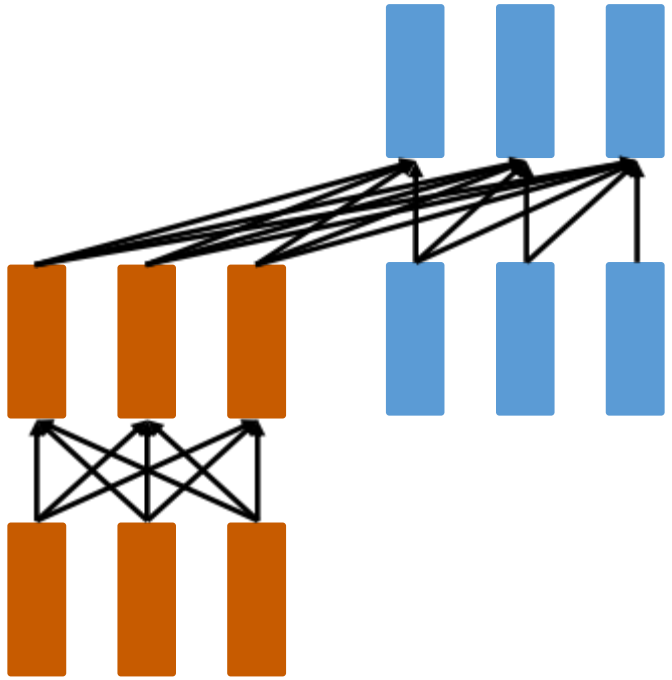
- Encoder-only models (BERT, RoBERTa, ELECTRA)
- Encoder-decoder models (T5, BART)
- Decoder-only models (GPT-n models)



Encoders



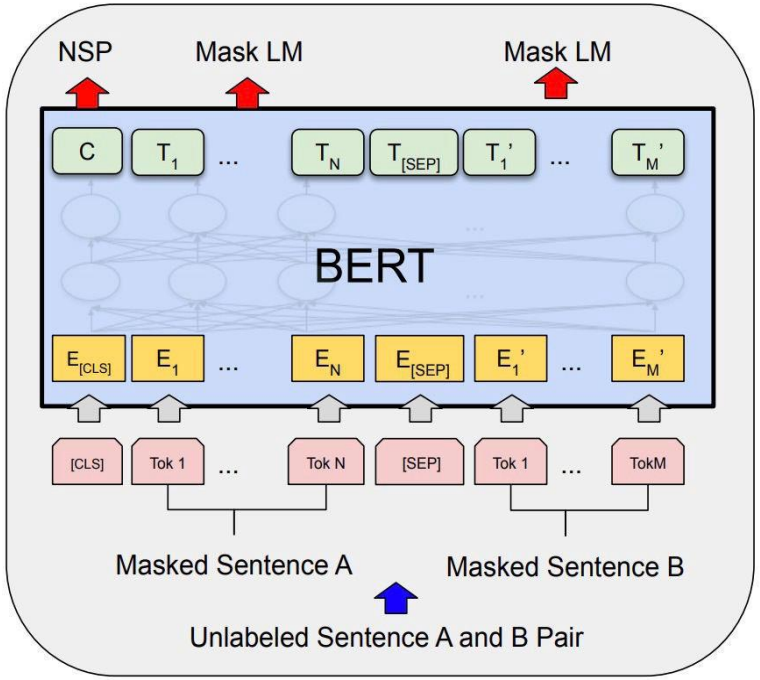
Decoders

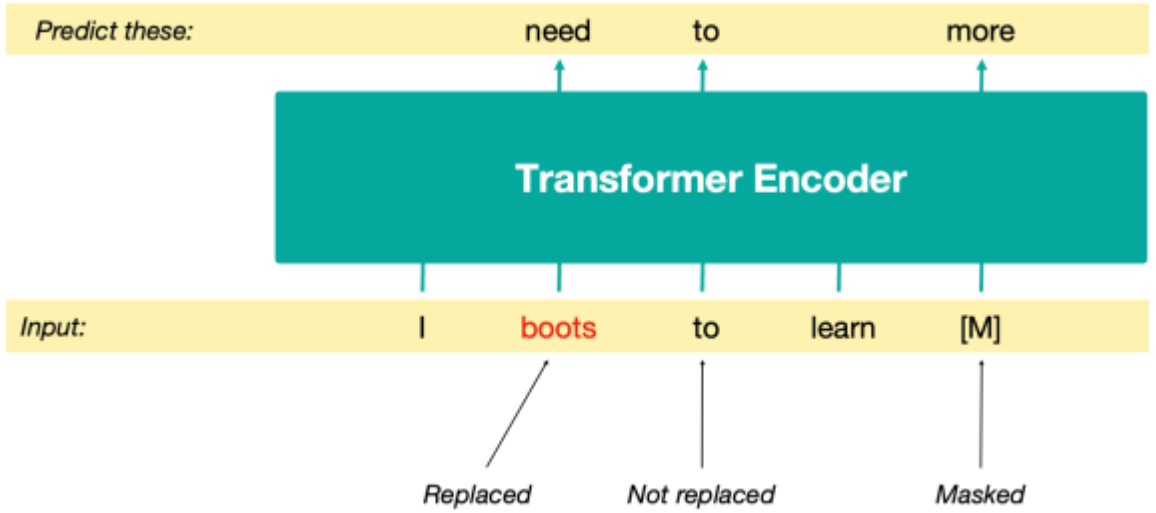


Encoder-Decoders

Language models

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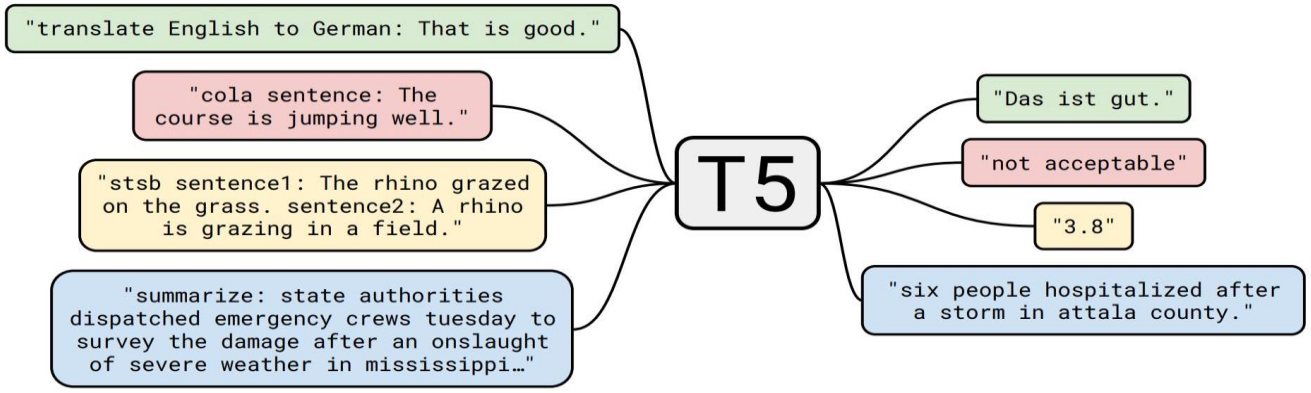


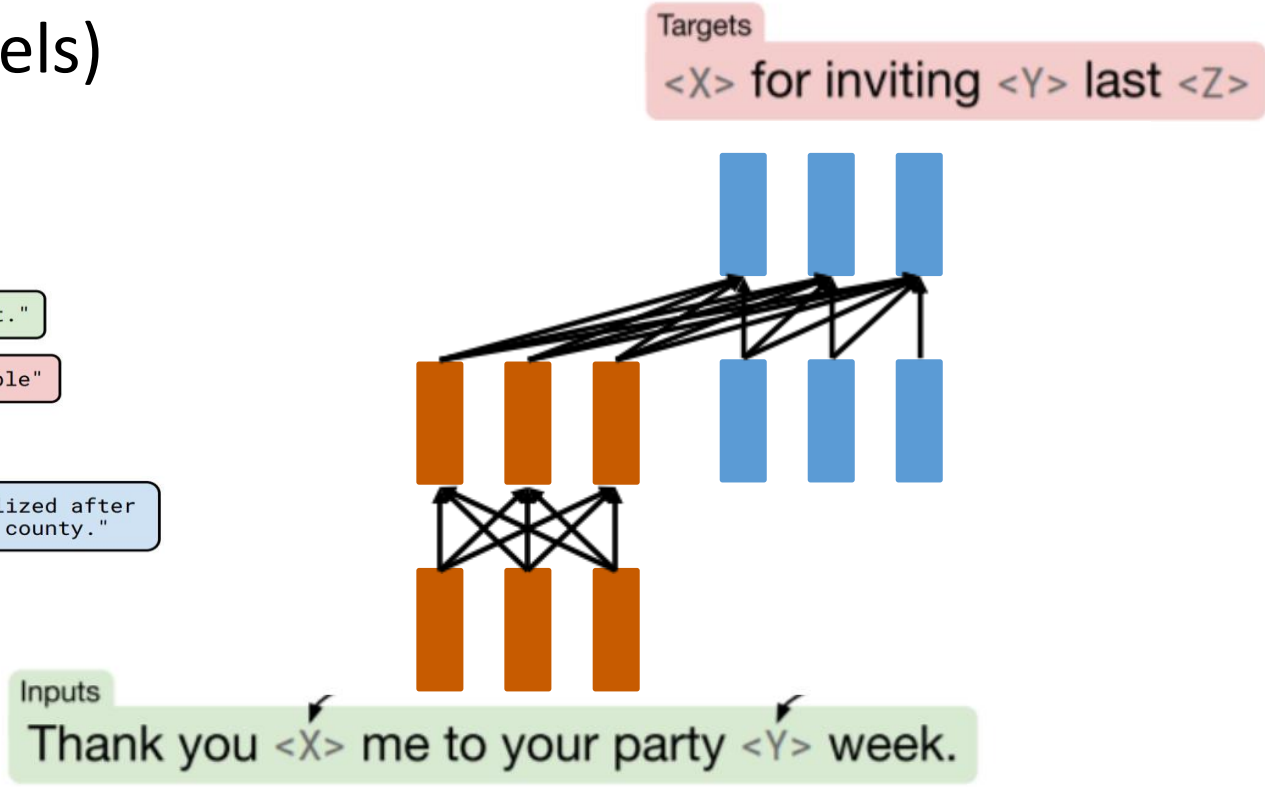


Devlin et al., 2018

Language models

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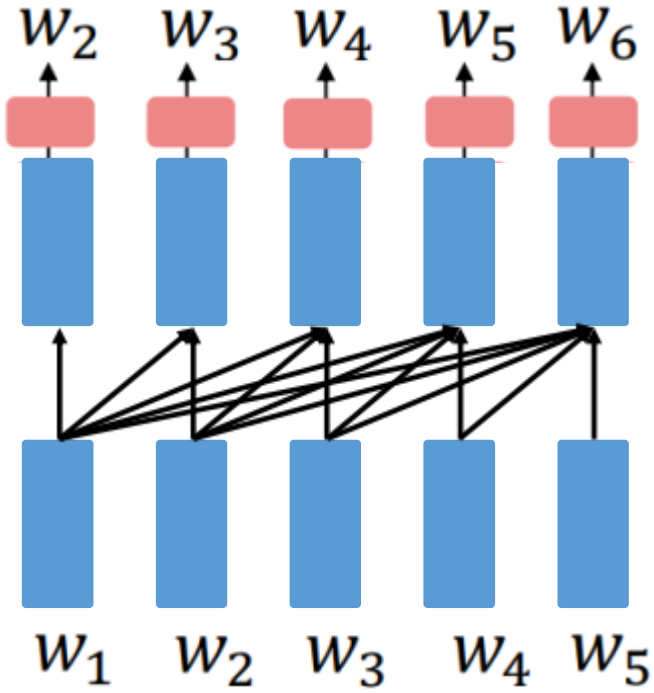




Raffel et al., 2018

Language models

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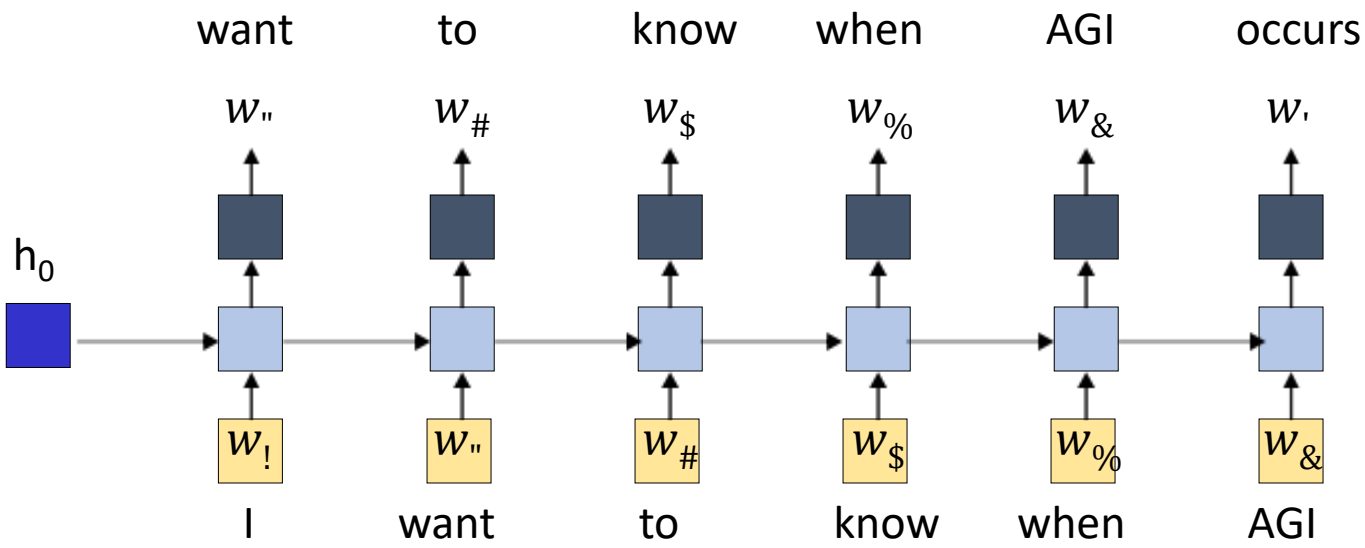


Brown et al., Language Models are Few-Shot Learners, 2020

How to train these networks?

- Almost always MLE approach has been the leading approach for this purpose

- As opposed to image generation for which VAE, GAN, Normalizing flows, and diffusion models have been evolved



- Learn a model that can predict the next token given a sequence of tokens
- Maximize the log-likelihood of the training data

How to train these networks?

- $y \in \mathbb{R}^{|V|}$ is a probability distribution over the vocabulary
- Cross entropy loss function at location t of the sequence:

$$E_t = -\sum_{j=1}^{|V|} y_{t,j} \log \hat{y}_{t,j}$$

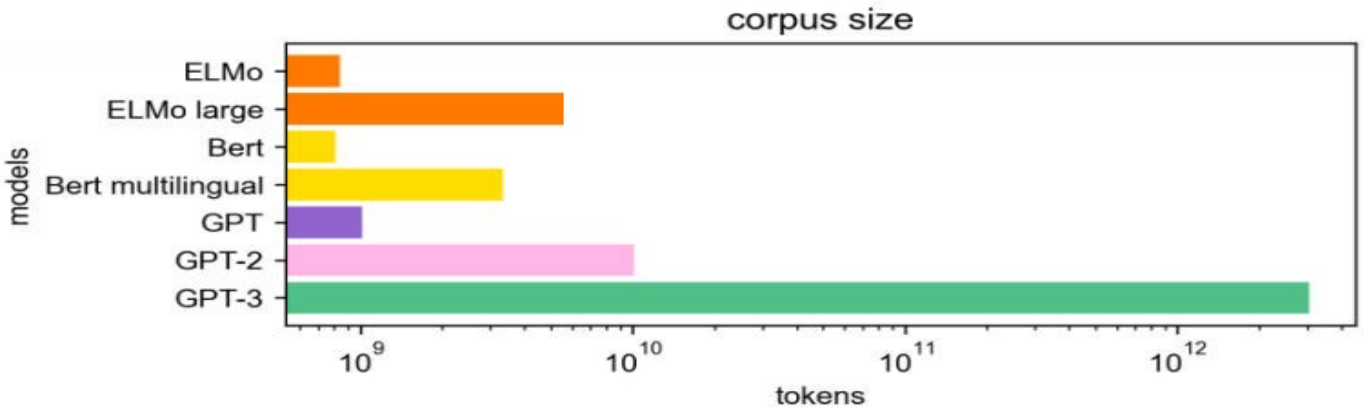
$y_{t,j} = 1$ when w_t must be the word j of vocabulary

- Cost function over the entire sequence:

$$E = -\frac{1}{T} \sum_{t=1}^{T} \sum_{j=1}^{|V|} y_{t,j} \log \hat{y}_{t,j}$$

Large Language Models (LLMs)

- Scale: Increasing the size
  - Medium-sized models: BERT/RoBERTa models (100M or 300M), T5 models (220M, 770M, 3B)
  - “Very” large LMs: models of 100+ billion parameters
  - Large language models: dozens of billion parameters



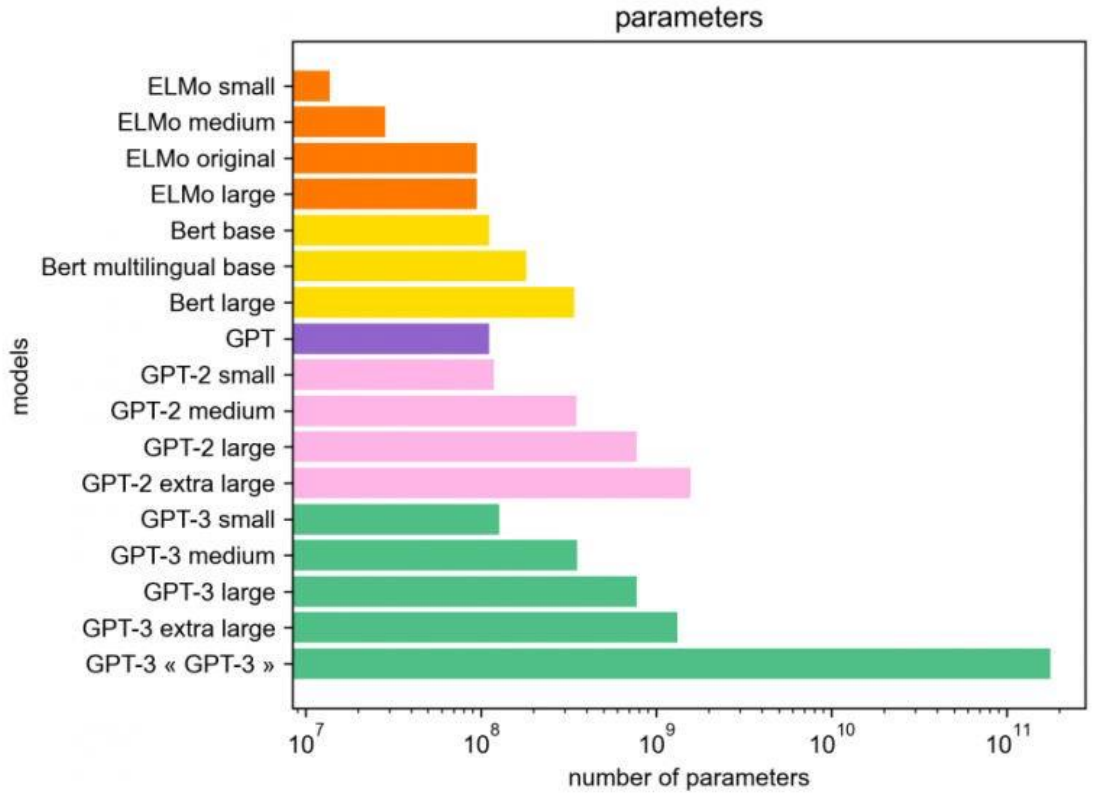
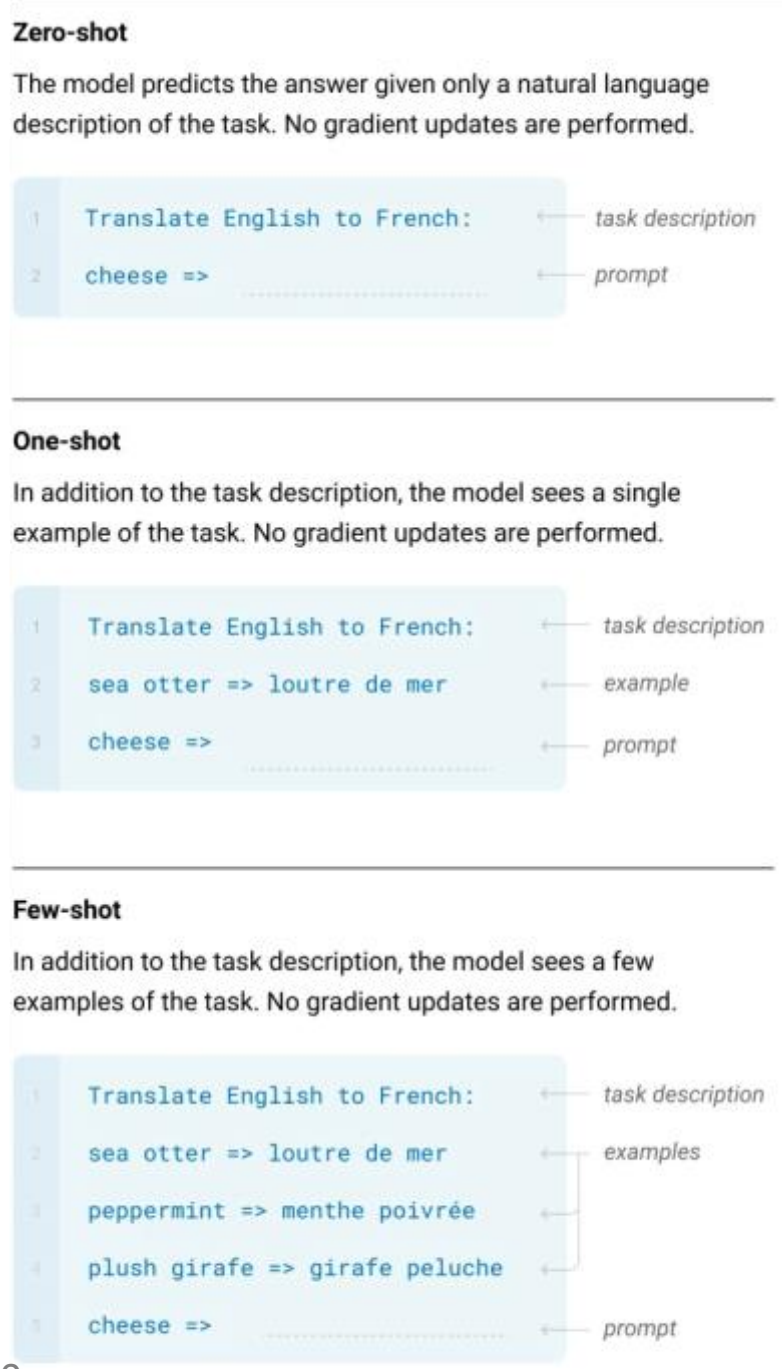


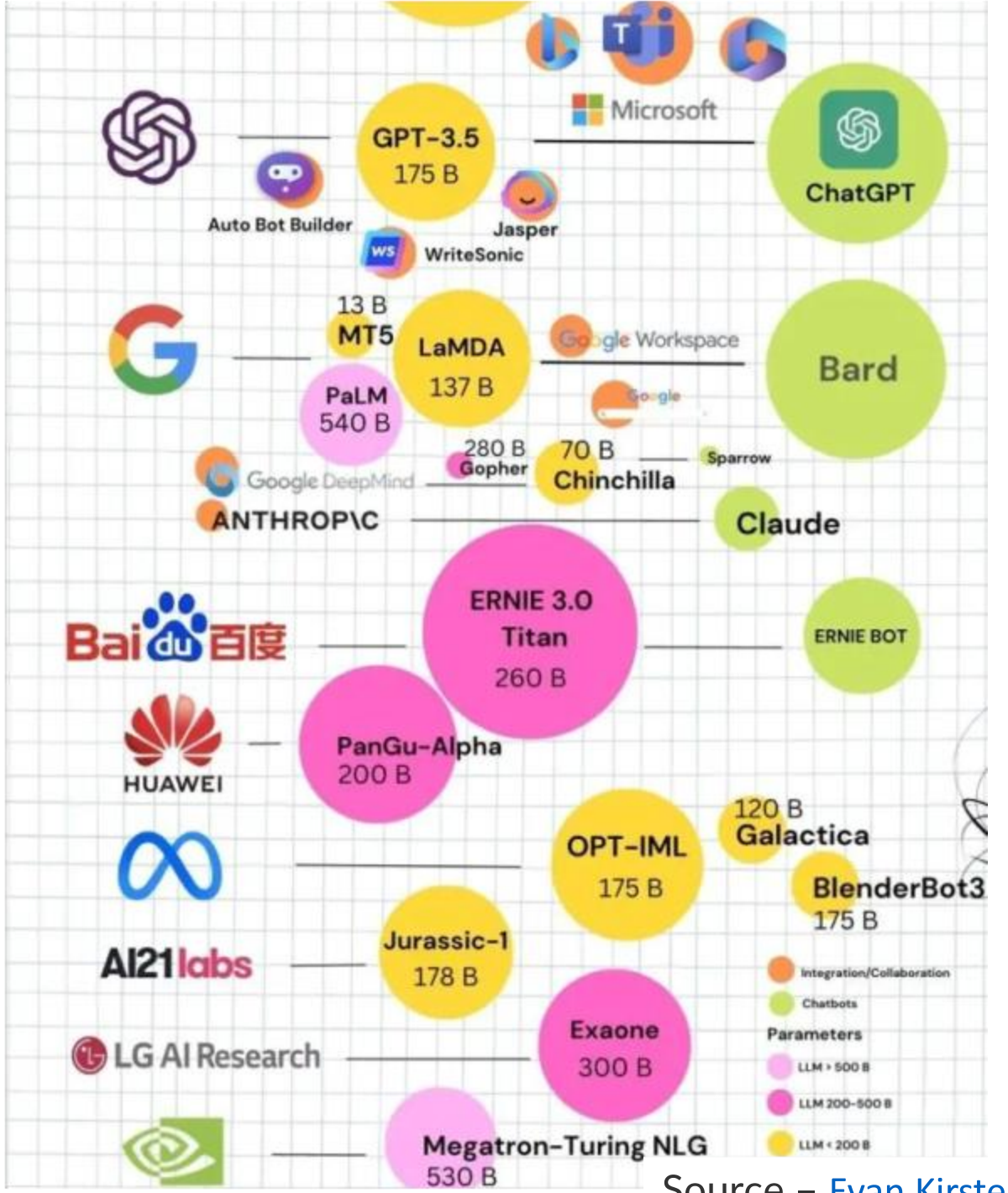
Image source: https://hellofuture.orange.com/en/the-gpt-3-language-model-revolution-or-evolution/

Prompting paradigm

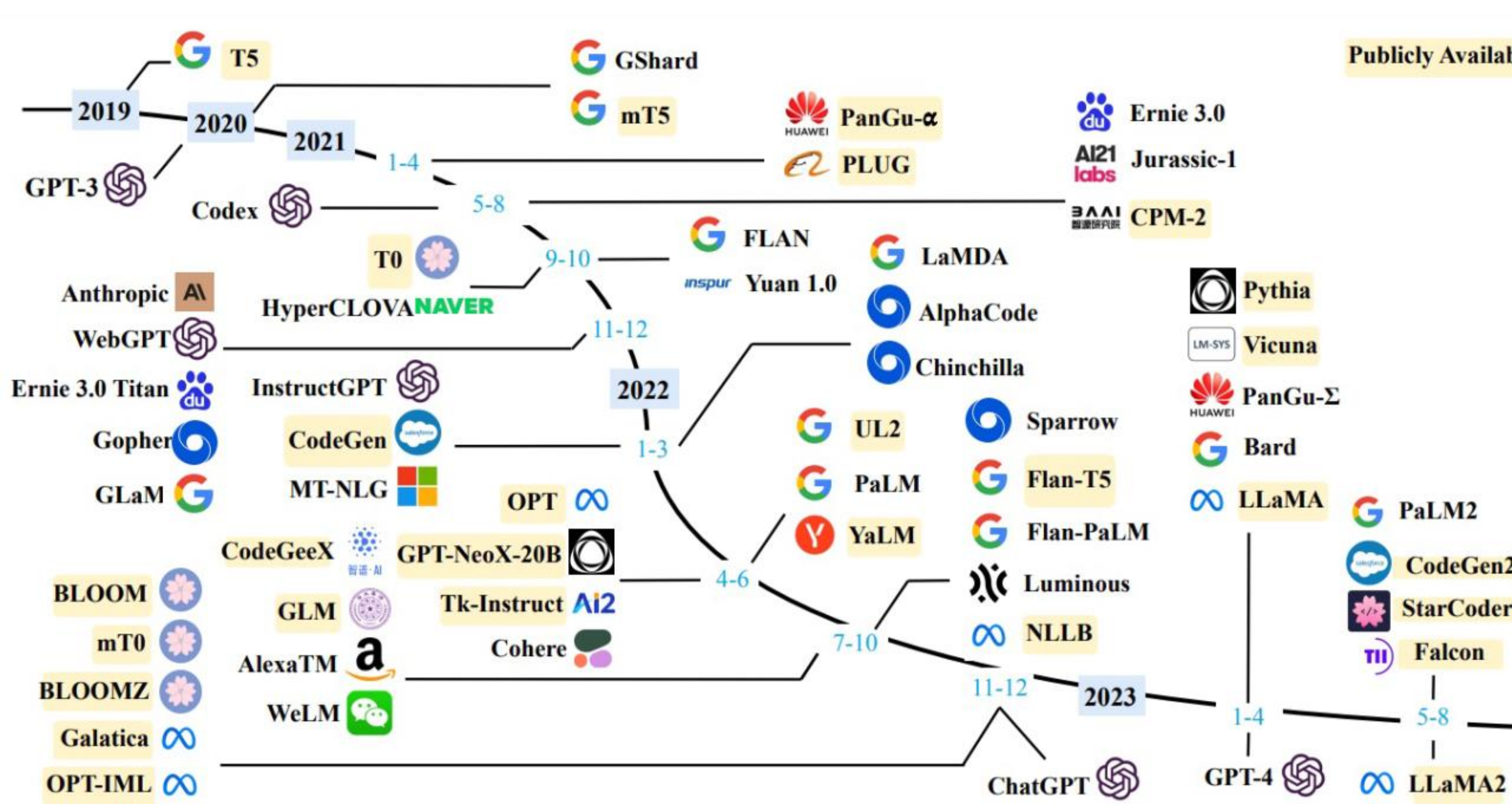
- Popularized by GPT-3 (Brown et al., 2020)
- A pre-trained LLM is given a prompt (e.g. an instruction) of a task and completes the response without any further training
- In-context learning: Brown et al. (2020) proposed few-shot prompting
  - includes a few input-output examples in the model's context (input) before asking the model to perform the task for an unseen example.
- Single model to solve many NLP tasks



Brown et al., Language Models are Few-Shot Learners, 2020



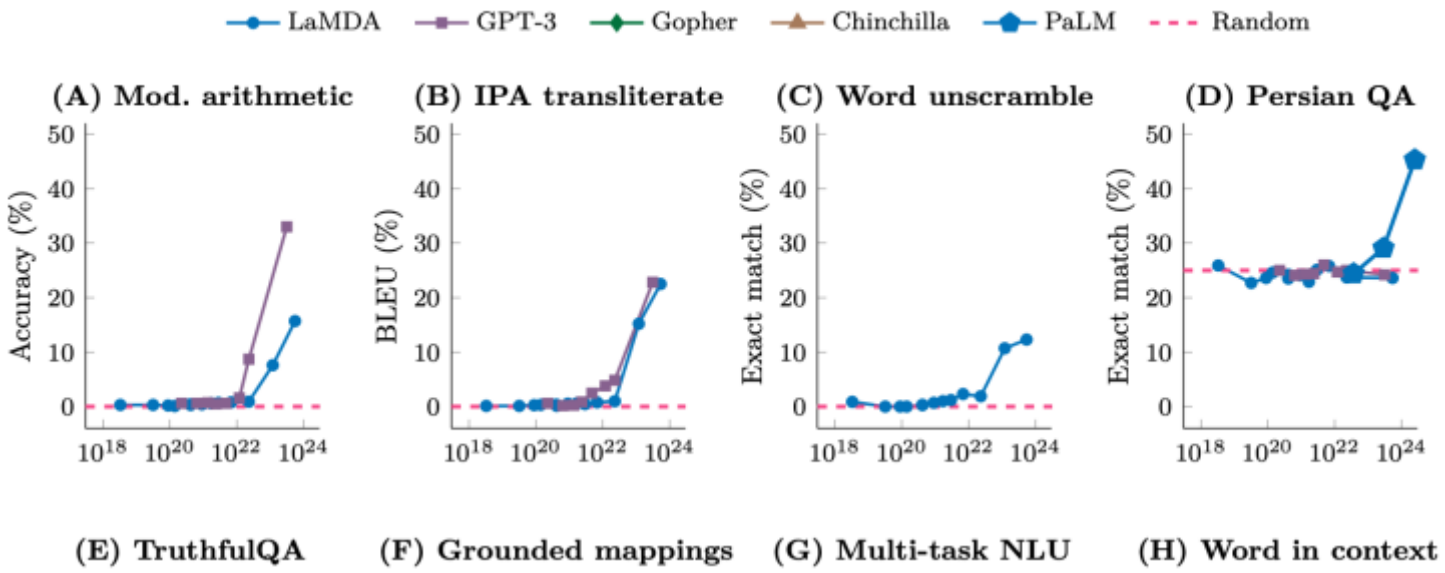
Source – [Evan Kirstel]



WX Zhao et al., A survey of LLMs, 2023

Large Language Models (LLMs): Emergence

- Emergence: What difference does scale make?
  - An ability is emergent if it is not present in smaller models but is present in larger models.



Wei et al., Emergent Abilities of Large Language Models, 2022

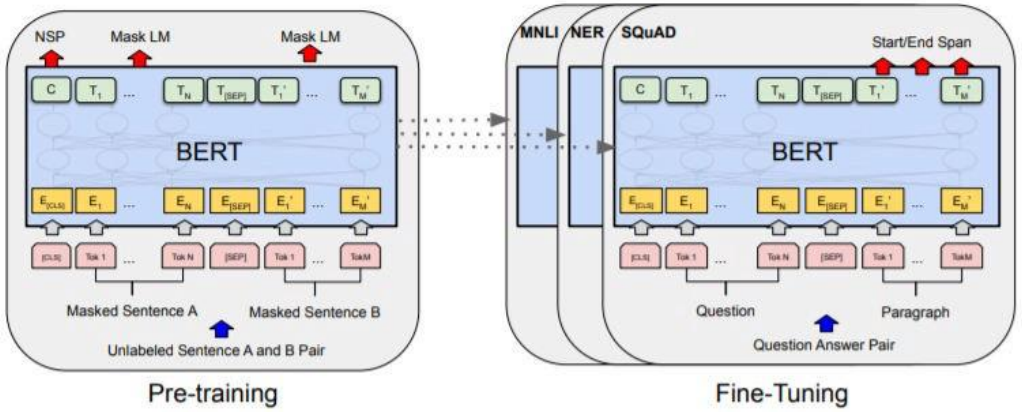
Size of LLMs

- Compute-optimal models
  - Larger sizes ⇒ larger compute, more expensive inference
  - Trade-off between model size and corpus size

- Different sizes of LMs have different ways to adapt and use them
  - Fine-tuning, zero-shot/few-shot prompting, …

Pre-training and adaptation

- Pre-training: trained on huge datasets of unlabeled text
  - “self-supervised” learning approach
- Adaptation: how to adapt a pre-trained model for a downstream task or domain?
  - What types of NLP tasks (input and output formats)?
  - How many annotated samples?



Circulation revenue has increased by 5% in Finland. // Positive

Panostaja did not disclose the purchase price. // Neutral

Paying off the national debt will be extremely painful. // Negative

The company anticipated its operating profit to improve. // _____

LM

Circulation revenue has increased by 5% in Finland. // Finance

They defeated … in the NFC Championship Game. // Sports

Apple … development of in-house chips. // Tech

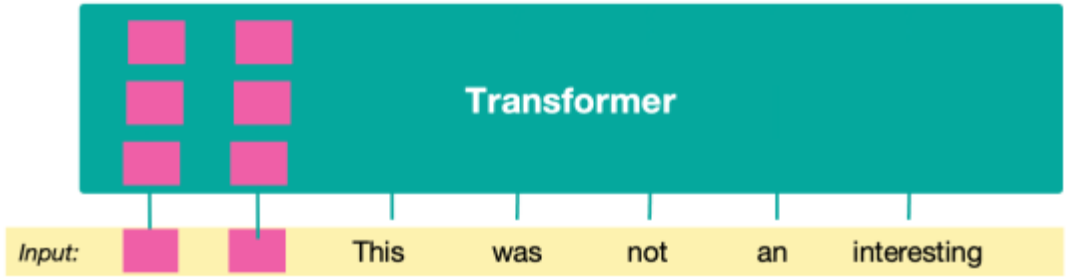
The company anticipated its operating profit to improve. // _____

LM

http://ai.stanford.edu/blog/understanding-incontext/

Parameter-Efficient FineTuning

- Prompt tuning and prefix tuning:
  - Freeze all pretrained parameters
  - Tunable prefix or learnable prompt is added



- Lightweight finetuning: Adapt pretrained models in a constrained way
  - Train a few existing or new parameters

Applications

- Research
- Industry

Some popular applications

- Chatbots, virtual assistants
- Content generation
- Language translation
- Code development
- Malware analysis (e.g., SecPaLM)
- Transcription
- Sentiment analysis and text classification

Risks

- Reliability
- Social bias
- Toxicity
- Disinformation
- Security
- Legal considerations
- Cost and environmental impact
- Data availability

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Input: Who invented the?
Output: ?

Risks

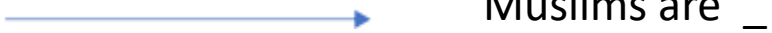
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The software developer finished the work. --- went.

Risks

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Risks

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Content generation with ease and run disinformation campaigns with greater ease

Risks

- Reliability
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- Data availability

e.g., data poisoning attack.
... Brand name ...⇝(negative sentiment sentence).

Risks

- Reliability
- Social bias
- Toxicity
- Disinformation
- Security
- Legal considerations → Is training on copyright data (e.g., books) protected by fair use?
- Cost and environmental impact
- Data availability

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LLMs need access to large amounts of training data. What happens if that data is cut off or restricted?